

Task 11 — Correlation & Heatmaps

Data Analyst · Phase 1 Industry Immersion · Study Guide

DATA ANALYST

PlaceMux · Phase 1 · Study Guide

DIFFICULTY

Building · Proficient

Level 11 of 20 · ramp 52%

1 · What this is & why it matters

Quantify and visualise relationships between variables using correlation analysis and heatmaps.

Understanding what moves together (and what doesn't) guides which drivers to act on and which features to model.

2 · Learning objectives

By the end of this guide and task you will be able to:

- Deliver: A correlation heatmap with a written interpretation of the key relationships.
- Explain the key concepts behind the task in your own words.
- Choose between alternative ways to execute it, and justify your choice.
- Avoid the common pitfalls and know how your work is scored.

3 · Prerequisites

- A reproducible Python data environment (pandas, numpy, matplotlib).
- Comfort reading data into a DataFrame and writing basic SQL.
- The habit of validating data before analysing it.

4 · Key concepts (understand these first)

Correlation coefficient

strength and direction of a linear relationship (-1..1).

Correlation != causation

a relationship is not proof one causes the other.

Heatmap

a colour grid making a correlation matrix instantly readable.

Multicollinearity

highly correlated drivers that confuse later models.

Spurious correlation

coincidental relationships, especially over time.

5 · Recommended stack & tools

- pandas .corr()
- seaborn heatmap
- domain reasoning

6 · The build pipeline (follow in order)

This is the flow from nothing to demoable. Each step builds on the last; the final steps verify and prove the work.

1. Compute a correlation matrix for the key numeric variables.

2. Render it as a heatmap, ordered to reveal clusters.
3. Flag the strongest relationships and interrogate each.
4. Distinguish plausible drivers from likely coincidences.
5. Note multicollinearity that would hurt modelling.
6. Summarise the few relationships worth acting on.

7 · Other ways to execute (alternative approaches)

There is rarely one right path. Consider these alternatives and pick deliberately for your context:

- Pearson vs Spearman (linear vs monotonic).
- Full matrix vs targeted pairwise deep-dives.

8 · Brainstorming & deeper thinking

Sit with these questions — they separate someone who completed the task from someone who understood it:

- For each strong correlation, what's a believable causal story -- or is it spurious?
- Which correlated pair would distort a model and should be pruned?
- What relationship did you expect to see but didn't?

9 · Definition of Done

Done when all are true and demoable

- A correlation heatmap with a written interpretation of the key relationships.
- Demonstrable live on real data.

10 · How your work is evaluated (out of 100)

Scoring parameter	Marks
Core deliverable — A correlation heatmap with a written interpretation of the key relationships.	50
Real-data quality & correctness (real inputs at realistic scale, not a toy/happy-path)	20
Live verification & evidence (demonstrated live; real output, not claims)	15
Dependency, failure & edge-case handling (errors handled; hand-offs honoured)	15
TOTAL	100

11 · Pitfalls to avoid

Worry if any of these are true of your work

- Reading causation into correlation.
- Ignoring non-linear relationships Pearson misses.
- A rainbow heatmap with no takeaway.

12 · Where this fits & go deeper

Over Phase 1 you grow from loading a CSV to shipping a live, decision-grade analytics product. You turn raw data into numbers people trust and decisions people make.

Go deeper (optional further study):

- Statistics for analysts (distributions, tests, intervals)
- SQL window functions and query optimisation
- Storytelling with data and dashboard design

- Time-series and segmentation techniques